Week 22
Constrained MDPs, Lagrangian Methods, CPO, FOCOPS, P3O

STUDY-RL — a self-paced Deep RL curriculum

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Learning objectives

Study companion. Read `teaching_guide.md` in this folder for step-by-step examples, an analogy with its limits, and self-checks. Then use `workbook.ipynb` to implement and vary the calculations.

Key Idea

By the end of this lecture you will be able to:

1. State Altman's Constrained MDP (CMDP) formulation as both a constrained optimisation in policy space and as a linear program in occupancy-measure space.
2. Derive the Lagrangian relaxation step-by-step, identify the saddle-point and write the primal-dual algorithm.
3. Implement PPO-Lagrangian end-to-end: dual advantage, projected multiplier update by projected descent on L, gradient-clipping, and the variance-control tricks (advantage normalisation, cost-scale normalisation).
4. Derive the CPO trust-region update from first principles, including the closed-form solution of the constrained quadratic program.
5. State and apply the FOCOPS closed-form non-parametric optimum and the P3O exact-penalty argument.
6. Define a coherent risk measure, derive Conditional Value-at-Risk, and read off CVaR from a distributional critic.
7. Diagnose the two canonical failure modes of Lagrangian RL (multiplier oscillation, cost-scale mismatch) and explain why PID-Lagrangian damps them.

1 Motivation & intuition

1.1 The hard problem nobody told you about

Up to week 21 we have been maximising expected return $J_r(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$. In practice this is rarely what we actually want.[1] The deployer almost always also cares about a side-quantity that must not blow up: an energy budget, a collision count, an unfairness metric, a maximum exposure to a particular candidate group, a clinical safety event rate.

In the unconstrained formulation those concerns get baked into the reward through a hand-tuned scalar: $r_t - \beta\, c_t$. This is brittle. The chosen β is almost always either too soft (constraint violated) or too harsh (reward collapses). Worse, the trade-off interpretation of β is *not* preserved during training — the agent has no view of the global constraint, only of the local linear combination.

The Constrained MDP framework promotes that scalar to a first-class variable $\lambda \geq 0$ that the algorithm *learns*, by formal duality. The result is a primal-dual algorithm that[2] solves *the right problem* — maximise reward subject to constraint.

1.2 Three mental models

Intuition. **The thermostat analogy.** λ is the price of cost. When the agent is overspending the budget, prices rise and the policy starts avoiding cost; when it is underspending, prices fall and the policy can afford to chase reward more aggressively. The equilibrium price is the marginal value of one extra unit of budget.

Intuition. **The market analogy.** Each (state, action) pair has two coupons: $r(s, a)$ and $c(s, a)$. The dual variable λ is the exchange rate between them. The optimal policy trades reward for cost at exactly that exchange rate; at any other rate, the budget is either wasted (slack constraint) or violated.

Intuition. **The KKT analogy.** At the saddle point of the Lagrangian we have stationarity (policy gradient zero), primal feasibility ($J_c \leq d$), dual feasibility ($\lambda \geq 0$), and complementary slackness ($\lambda(J_c - d) = 0$). Same KKT conditions you saw in convex analysis — but evaluated through Monte-Carlo estimates on policy-induced trajectory distributions.

[1] In recruiting, a "return" might be "successful hires" and a "cost" might be "rate of demographic-parity violation". Maximising one with zero regard for the other is an excellent way to build a discrimination-as-a-service product.

[2] This is exactly the same trick that turns Support-Vector-Machine training into the dual problem you saw in your ML course: a constrained quadratic program with KKT conditions, solved by alternating primal–dual updates.

2 Mathematical formulation

2.1 Altman's CMDP

Definition 2.1.1 (Constrained MDP)**.** A *constrained Markov Decision Process* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, c, \gamma, d)$ where $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ is a standard discounted MDP and $c : \mathcal{S} \times \mathcal{A} \to \mathbb{R}_{\geq 0}$ is a *cost function* with budget $d \geq 0$. The objective is

$$\max_{\pi} \; J_r(\pi) \triangleq \mathbb{E}_\pi\left[\sum_{t\geq 0} \gamma^t r(s_t, a_t)\right] \quad \text{subject to} \quad J_c(\pi) \triangleq \mathbb{E}_\pi\left[\sum_{t\geq 0} \gamma^t c(s_t, a_t)\right] \leq d.$$

In full generality there can be several cost channels with separate budgets; we keep one for clarity.

The LP view. For a finite CMDP, the *discounted state-action occupancy measure*

$$\mu^\pi(s, a) \;=\; (1-\gamma)\sum_{t\geq 0} \gamma^t \mathbb{P}_\pi(s_t = s, a_t = a)$$

satisfies $\sum_{s,a} \mu^\pi(s, a) = 1$ and the Bellman-flow constraint

$$\sum_a \mu(s, a) \;=\; (1-\gamma)\rho_0(s) + \gamma \sum_{s',a'} P(s \mid s', a')\, \mu(s', a').$$

The CMDP becomes a linear program over the polytope of valid occupancies:

$$\max_{\mu} \; \tfrac{1}{1-\gamma}\sum_{s,a} \mu(s, a)\, r(s, a) \;\text{ s.t. }\; \tfrac{1}{1-\gamma}\sum_{s,a} \mu(s, a)\, c(s, a) \leq d, \;\; \mu(s, a) \geq 0, \quad \text{Bellman flow.}$$

Because there are m scalar constraints, an optimal vertex of this polytope needs at most m randomisations beyond the basic deterministic feasible set — this is Altman, 1999's classical structural theorem. In practice it means the *optimal policy in a CMDP can be stochastic* even when the unconstrained optimum was deterministic — a fact you must internalise before you start debugging your safety algorithms.

Common pitfall

Determinism bias. Many practitioners default to deterministic policies because TD3/DDPG/SAC-tuned habits push that way. In a CMDP a constraint *may require randomisation at the optimal tight solution.* A deterministic policy can be feasible or even optimal, but restricting to deterministic policies can lose value or feasibility in some CMDPs.

2.2 The Lagrangian and its saddle-point

Form the (one-constraint) Lagrangian

$$L(\pi, \lambda) \;=\; J_r(\pi) - \lambda\,(J_c(\pi) - d), \qquad \lambda \geq 0. \tag{2.1}$$

Theorem 2.2.1 (Strong duality for finite CMDPs)**.** *For a finite discounted CMDP with $0 \le \gamma < 1$, fixed initial distribution, bounded rewards/costs, and a nonempty feasible occupancy polytope, strong duality holds:*

$$\max_{\pi} \min_{\lambda \ge 0} L(\pi, \lambda) \;=\; \min_{\lambda \ge 0} \max_{\pi} L(\pi, \lambda).$$

A primal-dual optimal pair $(\pi^\star, \lambda^\star)$ exists and the KKT conditions are

$$\begin{aligned} \textit{primal feas.} \quad & J_c(\pi^\star) \le d, \\ \textit{dual feas.} \quad & \lambda^\star \ge 0, \\ \textit{compl. slack.} \quad & \lambda^\star (J_c(\pi^\star) - d) = 0, \\ \textit{stationarity} \quad & \pi^\star \in \arg\max_{\pi} \; J_r(\pi) - \lambda^\star J_c(\pi). \end{aligned}$$

Proof sketch. The nonempty bounded occupancy polytope has an optimum. Finite-dimensional LP strong duality gives an optimal dual solution; strict feasibility of the cost inequality is not required for this LP result. Recover a stationary randomised policy by $\pi(a|s) = \mu(s,a)/\sum_b \mu(s,b)$ wherever the denominator is positive, and choose any action distribution at zero-occupancy states. The Bellman-flow equations ensure this policy induces the same occupancy. KKT then gives the displayed conditions. This argument optimises over all stationary randomised policies, not a restricted neural parameterisation.

□

Remark. For deep RL, the inner $\max_\pi$ at fixed λ is approximate (truncated gradient ascent on the surrogate). We do not get exact KKT in finite time; convergence requires additional assumptions on optimisation, estimation, feasibility, and timescales. A Robbins–Monro multiplier step-size schedule alone does not guarantee a saddle point for a neural policy.

2.3 Deriving the primal-dual update

Suppose π_θ is parametrised by θ. We want to ascend L in θ and descend L in λ:

$$\theta_{k+1} = \theta_k + \eta_\theta \, \nabla_\theta L(\pi_{\theta_k}, \lambda_k) = \theta_k + \eta_\theta \, (\nabla_\theta J_r(\pi_{\theta_k}) - \lambda_k \, \nabla_\theta J_c(\pi_{\theta_k})).$$

Both gradients are policy-gradient quantities:

$$\nabla_\theta J_r(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_t \gamma^t \nabla_\theta \log \pi_\theta(a_t|s_t) \, A_r^{\pi_\theta}(s_t, a_t) \right],$$

and analogously for cost. The combined gradient uses the *augmented advantage*

$$A^{\mathrm{tot}}(s,a) \;=\; A_r(s,a) - \lambda \, A_c(s,a), \tag{2.2}$$

which is the cleanest object to remember from this lecture.

For the multiplier we use projected gradient descent on L at fixed π. Since $\partial L / \partial \lambda = -(J_c - d)$, descent increases λ when cost exceeds the budget:

$$\lambda_{k+1} \;=\; \big[\, \lambda_k + \eta_\lambda \, (\hat{J}_c(\pi_{\theta_k}) - d) \,\big]_+. \tag{2.3}$$

$\hat{J}_c$ is the Monte-Carlo estimate from the current rollout.

Why we don't simply use $r - \beta c$. A fixed β is *exactly* the Lagrangian *at the optimum* $\beta = \lambda^\star$. Of course we do not know $\lambda^\star$ a priori. The primal-dual loop computes it for us.

2.4 The CPO trust-region update

Achiam et al. 2017 (arXiv:1705.10528) derive a cost-improvement bound and a practical constrained trust-region algorithm. Exact-model bounds and approximate sampled updates have different guarantees. The key inequality, derived from the standard performance-difference lemma (Kakade & Langford 2002):

$$J_c(\pi') - J_c(\pi) \;\le\; \tfrac{1}{1-\gamma}\,\mathbb{E}_{s\sim d^\pi, a\sim\pi'}\!\left[A_c^\pi(s,a)\right] \;+\; \tfrac{\sqrt{2}\,\gamma\,\epsilon_c^\pi}{(1-\gamma)^2}\,\sqrt{\bar{D}_{\mathrm{KL}}(\pi' \,\|\, \pi)}, \tag{2.4}$$

where $\epsilon_c^\pi \triangleq \max_s |\mathbb{E}_{a\sim\pi'}[A_c^\pi(s,a)]|$ is a worst-case cost-advantage scale. At each iteration CPO solves

$$\begin{aligned} \theta_{k+1} = \arg\max_\theta \;& g^\top(\theta - \theta_k) \\ \text{s.t.} \;& b^\top(\theta - \theta_k) + c \;\le\; 0, \\ & \tfrac{1}{2}(\theta-\theta_k)^\top H(\theta-\theta_k) \;\le\; \delta, \end{aligned} \tag{2.5}$$

where $g = \nabla_\theta J_r$, $b = \nabla_\theta J_c$ (cost gradient evaluated under the current policy), $c = J_c(\pi_k) - d$ (current constraint violation), H the Fisher information of π_θ at θ_k, and δ is the trust-region radius.

Theorem 2.4.1 (CPO closed-form update)**.** *Define $q = g^\top H^{-1} g$, $r_{bg} = g^\top H^{-1} b$, $s_b = b^\top H^{-1} b$. Assume H is positive definite and $s_b > 0$. The linearised constraint is feasible inside the trust region iff $c \le \sqrt{2\delta s_b}$. In the regular case with an active trust region, the optimal step is*

$$\theta_{k+1} - \theta_k \;=\; \tfrac{1}{\lambda_\star} H^{-1}(g - \nu_\star b),$$

where $\lambda_\star, \nu_\star$ solve the dual

$$\min_{\lambda\ge 0,\,\nu\ge 0} \;\; \tfrac{1}{2\lambda}(q - 2\nu r_{bg} + \nu^2 s_b) - \nu c + \lambda\delta, \qquad \lambda > 0.$$

For an infeasible linearised constraint ($c > 0$ and the trust region cannot satisfy it), CPO performs the recovery step $\theta_{k+1} = \theta_k - \sqrt{2\delta/s_b}\, H^{-1} b$, *which is the pure cost-reduction step inside the trust region.*

Proof. Form the Lagrangian $\mathcal{L}(\theta,\lambda,\nu) = -g^\top(\theta-\theta_k) + \nu(b^\top(\theta-\theta_k)+c) + \lambda(\tfrac{1}{2}(\theta-\theta_k)^\top H(\theta-\theta_k) - \delta)$. Stationarity in θ gives $(\theta-\theta_k) = H^{-1}(g-\nu b)/\lambda$. Substituting back gives the concave dual of the negative-reward minimisation. Negating that dual yields the displayed minimisation; this changes both the νc and $\lambda\delta$ signs. KKT classifies feasible vs. recovery regimes. □

CPO comes with a per-iteration bound: $J_c(\pi_{k+1}) \le d + $ (trust-region slack), which decays as $\sqrt{\delta}$.

Common pitfall

Fisher inversion. $H^{-1}g$ in CPO is computed by conjugate gradients on $Hv = g$, not by forming H explicitly. Each CG iteration costs one Hessian-vector product (forward + double backward through the log-policy). For 24–32 GB single-GPU budgets, choose CG iterations *at most* 10–20; beyond that you pay more than the policy step itself.

2.5 FOCOPS — first-order non-parametric optimum

Zhang, Vuong, Ross 2020 (arXiv:2002.06506). FOCOPS solves the same problem as CPO but in non-parametric policy space (working point: at fixed π_k, what is the optimal π'?), then projects.

Starting from the Lagrangian of the CPO QP *in non-parametric form*,

$$\max_{\pi'} \ \mathbb{E}_{s\sim d^{\pi_k}, a\sim\pi'}[A_r^{\pi_k}(s,a)] \ \text{ s.t. } \ \mathbb{E}[A_c^{\pi_k}(s,a)] + (1-\gamma)c \le 0, \ \ D_{\mathrm{KL}}(\pi'\|\pi_k) \le \delta,$$

and forming the Lagrangian $\mathcal{L}(\pi', \nu, \lambda)$, the KKT stationarity in $\pi'(a|s)$ gives

$$\log \pi'(a|s) = \log \pi_k(a|s) + \tfrac{1}{\lambda}\big(A_r^{\pi_k}(s,a) - \nu A_c^{\pi_k}(s,a)\big) - 1 + \mu(s),$$

where $\mu(s)$ is the multiplier on the row-normalisation $\sum_a \pi'(a|s) = 1$. Exponentiating and absorbing $\mu(s)$ into the normaliser:

$$\boxed{\pi^\star(a|s) \ \propto \ \pi_k(a|s)\, \exp\Big(\tfrac{1}{\lambda}\big(A_r^{\pi_k}(s,a) - \nu A_c^{\pi_k}(s,a)\big)\Big)\,.} \tag{2.6}$$

The multipliers λ, ν enter a coupled dual problem; they are not independent scalar root finds. The practical FOCOPS algorithm fixes a positive temperature λ, updates the nonnegative cost multiplier ν, and uses KL-based stopping/masking.

The projection minimises $D_{\mathrm{KL}}(\pi_\theta\|\pi^\star)$ over the parametric family (FOCOPS, Eq. 10). Reversing this KL would define a different projection, such as weighted maximum likelihood.

Why FOCOPS is friendlier than CPO on a single GPU. No Fisher-vector products, no conjugate gradients — both phases are first-order. The projection and sampled KL controls remain approximate. An average-KL condition is not a worst-case per-state guarantee, and this distinction also matters when interpreting practical CPO.

2.6 P3O — penalised PPO

Zhang et al. 2022 (arXiv:2205.11814) use an exact-penalty approach. With S_r^{clip} denoting a reward surrogate to *maximise*, its sign convention is

$$S_{\mathrm{pen}}(\theta) = S_r^{\mathrm{clip}}(\theta) - \kappa[J_c(\pi_\theta) - d]_+. \tag{2.7}$$

A minimising optimiser instead receives $-S_r^{\mathrm{clip}} + \kappa[J_c - d]_+$. Real P3O uses sampled cost surrogates rather than backpropagating through an observed scalar episode cost.

Theorem 2.6.1 (Exact penalty for the ideal constrained problem)**.** *Suppose strong duality holds and a finite optimal multiplier $\lambda^\star$ exists. For $\kappa > \lambda^\star$, every global maximiser of $J_r(\pi) - \kappa[J_c(\pi) - d]_+$ is feasible and constrained-optimal.*

Proof. A feasible optimum attains $J_r^\star$. For a violating policy with $v = J_c - d > 0$, dual optimality gives $J_r - \kappa v \le J_r^\star - (\kappa - \lambda^\star)v < J_r^\star$. A feasible suboptimal policy also cannot maximise the penalised objective. □

This proof concerns exact objectives and global optimisation. It does not make every local optimum of a neural PPO surrogate feasible. At the hinge, a valid subgradient may be chosen; below the budget the penalty is zero.

2.7 Risk-sensitive primer: CVaR and distributional critics

Definition 2.7.1 (Conditional Value-at-Risk)**.** For a random variable G with c.d.f. F_G and level $\alpha \in (0, 1]$,

$$\mathrm{VaR}_\alpha(G) = \inf\{g : F_G(g) \ge \alpha\}, \qquad \mathrm{CVaR}_\alpha^{\mathrm{lower}}(G) = \tfrac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_u(G)\, du.$$

This lower-tail return functional captures the worst α fraction of outcomes, fractionally weighting an atom at the quantile when necessary. For continuous distributions it equals the conditional mean below the quantile. The corresponding upper-tail loss risk measure is coherent; do not transfer the sub-additivity sign unchanged to a lower-tail reward utility. Here $\alpha \to 1$ recovers $\mathbb{E}[G]$, $\alpha \to 0$ recovers worst-case.

With a distributional critic $Z^\pi(s,a)$ (C51, QR-DQN, IQN — introduced in Week 6), CVaR becomes a *post-hoc* computation on the empirical quantile function. The risk-sensitive greedy action is

$$a^\star(s) \;=\; \arg\max_a \; \mathrm{CVaR}_\alpha\big(Z(s,a)\big).$$

For a cost variable C, the worst α fraction is the *upper* tail, $\alpha^{-1}\int_{1-\alpha}^{1} \mathrm{VaR}_u(C)\,du$. Week 24's shipped implementation uses a mean parity statistic; it does not implement a CVaR cost constraint.

Intuition. Expected cost describes the average. Upper-tail CVaR at tail mass 0.05 describes the worst 5% of cost outcomes. Lower-tail CVaR of cost instead selects the smallest costs and would answer the wrong safety question.

3 Algorithms

3.1 PPO-Lagrangian pseudocode

Algorithm: PPO-Lagrangian (one update)

Inputs: policy π_θ, value heads V_r, V_c, multiplier λ, budget d.

1. Collect T steps of π_θ rollout, storing $\{(s_t, a_t, r_t, c_t, V_r(s_t), V_c(s_t), \log \pi_\theta(a_t|s_t))\}$.
2. Compute GAE advantages A_r on reward, A_c on cost (with respective bootstrap targets). *Normalise* A_r to zero mean / unit std. *Do not* normalise A_c — the multiplier needs to see the raw cost scale.
3. Combine: $A^{\text{tot}} = \dfrac{A_r - \lambda A_c}{1 + \lambda}$. The $1 + \lambda$ denominator keeps the coefficients on the two advantages bounded as λ grows; this is not a bound on arbitrary neural gradients.
4. PPO update on π_θ with A^{tot}, clip ratio ϵ, K epochs, minibatches of size M. Train V_r, V_c by MSE to GAE returns.
5. Dual update: $\lambda \leftarrow \left[\lambda + \eta_\lambda(\hat{J}_c - d)\right]_+$, clipped to $[0, \lambda_{\max}]$.

Complexity. Each update is $\mathcal{O}(T \cdot K)$ forward passes plus the same many backward passes. The multiplier update is $\mathcal{O}(1)$. Storage is $\mathcal{O}(T)$ for the rollout buffer.

3.2 A numerical budget and multiplier check

Consider one state that transitions back to itself. Action 1 has reward and cost 1; action 0 has reward and cost 0. A stationary policy takes action 1 with probability p, so $J_r = J_c = p/(1-\gamma)$. With $\gamma = 0.9$ and $d = 0.5$, feasibility requires $p \le 0.05$, not $p \le 0.5$. At $p = 0.5$, $J_c = 5$. Starting from $\lambda = 0$ with step size 0.1, one dual update gives $\lambda' = 0.1(5 - 0.5) = 0.45$. This arithmetic checks the sign and units; it does not prove convergence after a fixed number of updates. The constrained optimum has $p = 0.05$ and reward 0.5.

3.3 CPO pseudocode

Algorithm: CPO (one update)

Inputs: π_{θ_k}, trust-region δ, budget d.

1. Rollout, estimate A_r, A_c, current violation $c = J_c - d$.
2. Build $g = \widehat{\nabla} J_r$, $b = \widehat{\nabla} J_c$ via score function with advantages.
3. Compute $H^{-1}g$ and $H^{-1}b$ by conjugate gradients on the Fisher–vector product $v \mapsto \widehat{\nabla^2}_\theta \mathrm{KL} \cdot v$.
4. Solve the dual of eq. (2.5); if feasible take the trust-region step, else recovery step.
5. Backtracking line search: shrink the step until both the KL constraint and an estimate of the cost constraint are satisfied.

3.4 FOCOPS pseudocode

Algorithm: FOCOPS (one update)

1. Rollout, estimate A_r, A_c.
2. Use a fixed positive temperature λ and update ν from the cost violation, as in practical FOCOPS; the ideal dual is coupled.
3. Form the target $\pi^\star$ via eq. (2.6) (only at sampled states).
4. Project onto π_θ by SGD on $\mathrm{KL}(\pi_\theta \| \pi^\star)$.

3.5 P3O pseudocode

Algorithm: P3O (one update)

1. Rollout, compute the PPO surrogate L_r and the cost estimate $\hat{J}_c$.
2. Compute the hinge $H = \max(0, \hat{J}_c - d)$.
3. Minimise $-L_r + \kappa H$, where L_r is the reward surrogate to maximise. Construct a differentiable score-function cost surrogate; an observed scalar cost alone has no policy gradient.

Common pitfall

Cost-scale mismatch. Reward and cost advantages have different units. If scaling costs by a fixed $s > 0$, also scale the budget by s: $c' = c/s$, $d' = d/s$. To preserve $r - \lambda c$, use $\lambda' = s\lambda$. Untracked running normalisation changes the constraint or its effective price. Normalising reward advantages also changes that price, so document this practical heuristic rather than interpreting the resulting multiplier as an exact economic shadow price.

4 Implementation walkthrough

We implement PPO-Lagrangian on the `LavaGridWorld` environment in `project/env.py`. The architecture is intentionally minimal: 2 MLPs (one trunk for policy, one for the two value heads) of 64-64 hidden units, one-hot state observations on a 7×7 grid.

4.1 Environment: a constrained lava-grid

```python
@dataclass
class LavaGridWorld:
    rows: int = 7; cols: int = 7
    start = (0, 0); goal = (6, 6)
    lava = ((1,2),(1,3),(2,2),(2,3),(4,3),(4,4),(5,3),(5,4))
    max_steps: int = 80
    slip_prob: float = 0.05
    step_reward: float = -0.01
    goal_reward: float = 1.0
    lava_cost: float = 1.0
    # step(action) -> (next_idx, reward, cost, done, info)
```

Cost is +1 *per visit* to a lava cell — the agent can recover (it does not terminate). This is the standard CMDP setup; the cost is just another scalar channel.

4.2 Networks

Two separate trunks for actor and critic, mostly to keep gradient-graph hygiene clean across PyTorch's autograd. The actor outputs a categorical over 4 actions; the critic has *two heads*, V_r and V_c, with weights tied only through their shared trunk.

```python
class ActorCritic(nn.Module):
    def forward(self, obs):
        h_pi = self.pi_trunk(obs); h_v = self.v_trunk(obs)
        return self.pi_head(h_pi), self.v_head(h_v).squeeze(-1),
        ↪  self.vc_head(h_v).squeeze(-1)
```

4.3 GAE on two streams

```python
def compute_gae(rews, vals, dones, gamma, lam, last_val=0.0):
    T = len(rews); adv = np.zeros(T, dtype=np.float32)
    gae = 0.0; next_val = last_val
    for t in reversed(range(T)):
        nonterm = 1.0 - float(dones[t])
        delta = rews[t] + gamma*next_val*nonterm - vals[t]
        gae = delta + gamma*lam*nonterm*gae
```

```
        adv[t] = gae; next_val = vals[t]
    return adv, adv + np.asarray(vals, dtype=np.float32)
```

The same routine is called twice per update: once with (r_t, V_r, done) and once with (c_t, V_c, done). Cost advantages are *not* normalised — λ has to see the cost scale.

4.4 The combined PPO loss

```
adv_total = (adv_r - self.lam * adv_c) / (1.0 + self.lam)
ratio = torch.exp(logp - old_logps)
clip_adv = torch.clamp(ratio, 1 - eps, 1 + eps) * adv_total
pi_loss = -(torch.min(ratio * adv_total, clip_adv)).mean()
```

The $(1 + \lambda)$ denominator forms coefficients $1/(1 + \lambda)$ and $\lambda/(1 + \lambda)$, preventing the relative scale from growing linearly with the multiplier for fixed advantages. This does not bound the policy Jacobian, variance, or training instability.

4.5 Dual update

```
mean_cost = float(np.mean(ep_costs))
self.lam = float(np.clip(self.lam + self.cfg.lam_lr * (mean_cost -
↪  self.cfg.cost_limit),
                         0.0, self.cfg.lam_max))
```

This is the simplest possible projected-ascent rule. PID-Lagrangian (exercise 8) replaces it with a PID controller; FOCOPS replaces it with a different first-order update with a fixed KL temperature and adaptive cost multiplier. The interface (λ as a Python float on the agent) stays the same.

Common pitfall

Episode boundaries in cost estimates. $\hat{J}_c$ must be the *discounted* cost over completed episodes, not the per-step instant cost. Using per-step costs collapses the budget interpretation. The `train.py` loop accumulates $\sum_t \gamma^t c_t$ explicitly and resets at every done.

5 Empirical results

5.1 Statistics bridge: An average-cost budget is different from a tail restriction

Example 5.1.1 (Original statistical teaching fixture)**.** A fixed policy has episode cost 0 with probability 0.75 and cost 4 with probability 0.25. The expected-cost budget is 1; this is a fully specified toy distribution.

Expected cost is $0.75(0) + 0.25(4) = 1$, exactly the budget. Cost variance is $0.25(16) - 1^2 = 3$. The worst quarter of episodes all cost 4, so the upper-tail average over that quarter is 4 even though expected cost equals 1.

Expected-cost feasibility does not impose a cost limit on every episode or control a separately defined tail metric. In a learned policy, estimating the mean and satisfying the population budget are additional distinct questions. An optimizer or multiplier trace does not certify either by itself.

Check a changed assumption. Change the cost-4 probability to 0.1. Calculate mean cost and the worst-quarter average, including the zero-cost mass needed to fill that quarter.

Check. Mean cost becomes 0.4. The upper quarter contains probability mass 0.1 at cost 4 and 0.15 at cost 0, giving tail average $0.4/0.25 = 1.6$. Fractional tail mass matters.

Statistical foundation. Siegrist § § 4.1, 4.3, PDF pp.439, 460. Section source: https://stats.libretexts.org/@go/page/10158. The course statistics companion records the supplied editions and one-based PDF pages. This fixture is not a newly measured project result.

5.2 Setup

We train PPO-Lagrangian for 80 updates of 512 steps each on the lava grid ($\gamma = 0.99$, GAE $\lambda_{\text{GAE}} = 0.95$, clip 0.2, 4 PPO epochs, minibatch 128, $\eta_\lambda = 0.05$, budget $d = 1.5$). All runs use seed 0 and reproduce in ∼30 s on a single CPU core.

5.3 Results

5.4 Reading the curves

The cost curve clearly shows the dual mechanism: initial overshoot, multiplier-driven correction, eventual convergence to a slack value (below budget). Compare the unconstrained baseline in fig. 5.3, which never sees the constraint signal and stays high on cost forever — just as the theory predicts.

Safety angle

The CMDP is the smallest formalism in which "safety" is a first-class object rather than a hyperparameter. But the formalism is not magical:

- **Constraint specification gaming.** The agent only respects the constraint you wrote,

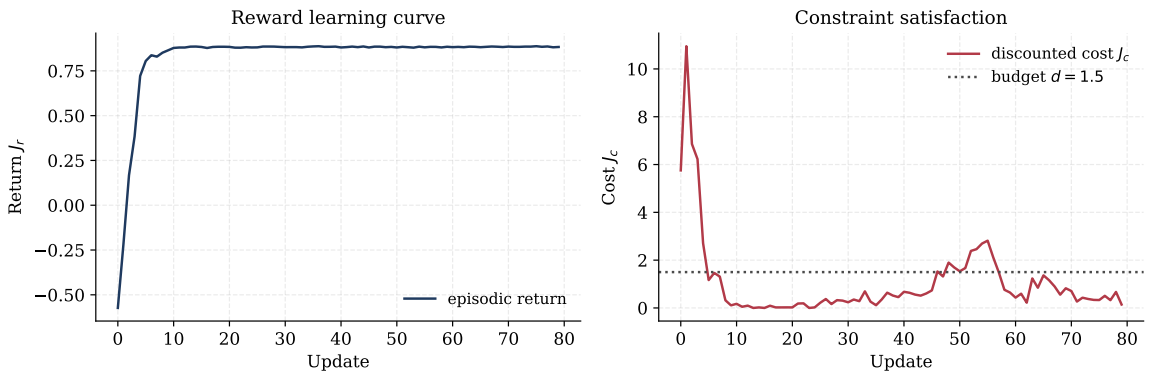


Figure 5.1: Left: episodic reward J_r. Right: discounted cost J_c tracked against the budget $d = 1.5$ (dotted). The policy starts above budget, the multiplier spikes, and the agent learns a route that hits the goal while skirting lava.

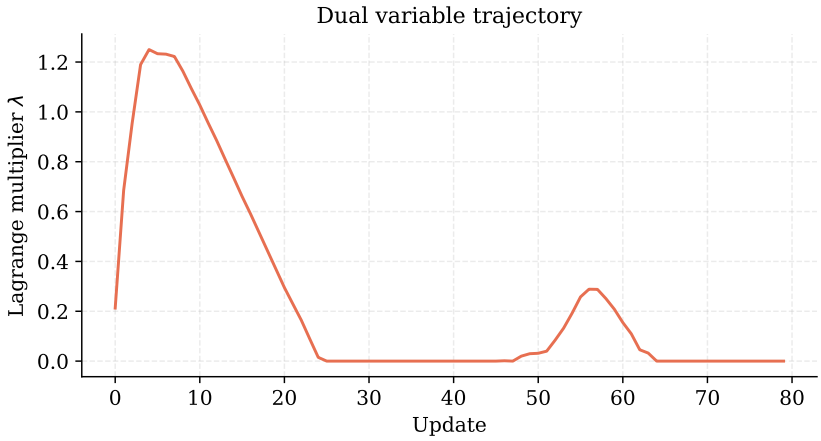


Figure 5.2: Lagrange multiplier λ over training. The sharp early spike is the system pricing the constraint violation; it then settles near 0 once the agent's cost falls below budget (complementary slackness in action).

not the one you meant. If c is "visits to a lava cell" and the lava also melts the agent, the agent may find a policy that minimises lava-visits while never actually completing the task — effectively gaming the constraint. Recommended defence: *multiple* cost channels, each capturing one slice of the intended safety property (rate, severity, duration).

- **Asymptotic only.** The Lagrangian guarantees feasibility at the saddle point. During training there is no guarantee. For real deployment you must wrap the policy in a runtime safety filter (*week 23*: shielding, CBFs).
- **Multiplier oscillation under cost noise.** The dual update is a noisy projected descent on L; when $\hat{J}_c$ has high variance, λ can oscillate wildly, driving the policy back and forth between greedy and risk-averse modes. PID-Lagrangian (Stooke et al. 2020, arXiv:2007.03964) damps these oscillations.

References: Ray, Achiam, Amodei 2019, author-published Safety Gym report, "Benchmarking safe exploration in deep RL"; Stooke, Achiam, Abbeel 2020, arXiv:2007.03964, "Responsive safety in RL by PID Lagrangian methods".

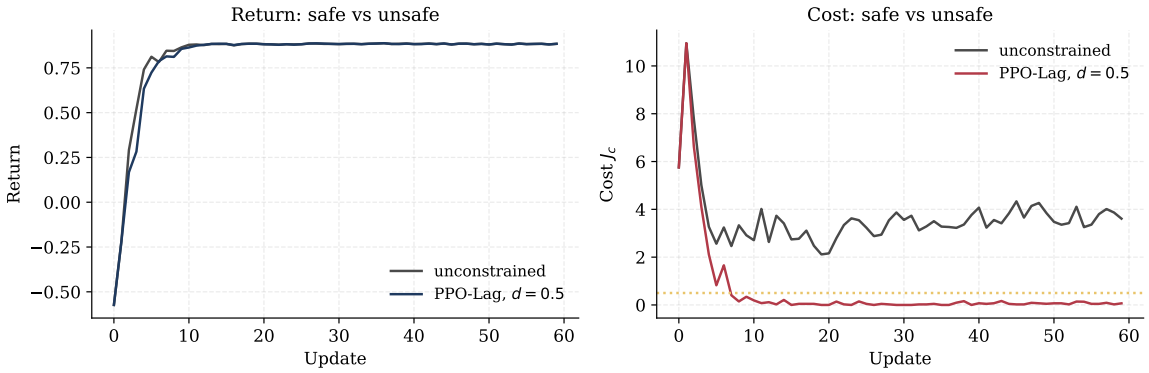


Figure 5.3: Strict-budget PPO-Lag ($d = 0.5$) vs. unconstrained PPO (effectively $d = \infty$). The unconstrained policy converges to the shortest path, which goes *through* lava and incurs high cost; the constrained policy detours, with a small return penalty.

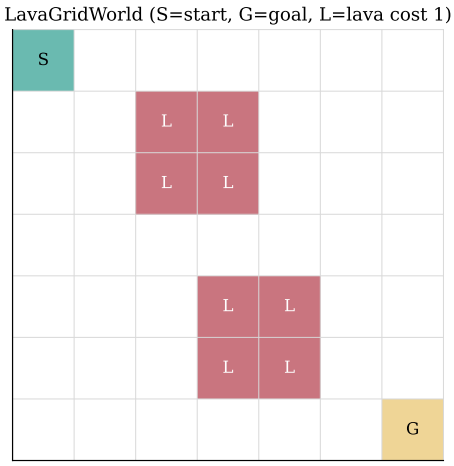


Figure 5.4: The lava gridworld used throughout this chapter.

Matching application

For the synthetic matching exercise, define the population parity gap as $|\Pr_\pi(a = 1 \mid g = 0) - \Pr_\pi(a = 1 \mid g = 1)|$. Both conditional rates need eligible-population denominators and a stated sampling distribution. With fixed exogenous positive group frequencies, bound the signed gap above and below by d using two linear constraints in action probabilities. The quantity $|\nVdash[g(a) = 1] - \rho|$ is not this parity gap: at $\rho = 1/2$ it is always $1/2$, regardless of the policy. Week 24 demonstrates a differentiable finite-batch probability-gap surrogate. It is a synthetic exercise, not a production fairness guarantee. The multiplier is a learned feedback variable; it is not by itself an audit of eligibility, outcomes, uncertainty, or legal obligations.

For synthetic insurance lead routing the analogue is: r = expected gross-written-premium contribution; c = saturation of any one broker beyond their capacity quota; d = broker-quota slack policy. PPO-Lag respects the broker fairness constraint without manual scalarisation.

6 Exercises

See `workbook.ipynb` for graded exercises with hints and an auto-graded reference. A summary:

- **Ex. 1 ($\star$):** one-step Lagrangian dual update.
- **Ex. 2 ($\star$):** P3O hinge subgradient.
- **Ex. 3 ($\star\star$):** GAE for the cost stream.
- **Ex. 4 ($\star\star$):** FOCOPS closed-form non-parametric optimal.
- **Ex. 5 ($\star\star$):** empirical CVaR.
- **Ex. 6 ($\star\star$):** train PPO-Lag, verify constraint satisfaction.
- **Ex. 7 ($\star\star\star$):** compare constrained vs. unconstrained; explain the gap.
- **Ex. 8 ($\star\star\star$):** PID-Lagrangian one-step update.

7 Further reading

The bibliography for this week is in `extended_reading.pdf`:

- Altman, *Constrained Markov Decision Processes*, CRC 1999 — the canonical reference.
- Ray, Achiam, Amodei, "Benchmarking safe exploration in deep RL", author-published Safety Gym report (2019).
- Achiam, Held, Tamar, Abbeel, "Constrained Policy Optimization", arXiv:1705.10528 (2017).
- Zhang, Vuong, Ross, "FOCOPS", arXiv:2002.06506 (2020).
- Stooke, Achiam, Abbeel, "Responsive safety by PID Lagrangian", arXiv:2007.03964 (2020).
- Zhang, Vuong, Ross, "P3O", arXiv:2205.11814 (2022).

Sourced from `research/safety_rl_sota.md`, sections 1 and 3.